

Applied Cloud Computing

Assignment 1 : Part 1: Amazon EC2 Intro (updated Sept. 2025)

In this assignment, you will create and launch an EC2 instance, log on using SSH and test basic HTTP Service.

Here are the steps:

Step 1. You should have signed up an AWS account (see lecture note)

Step 2. Sign in [AWS Management Console](#)

Note : After signing in, check which **region** you are in.

You can see this at the top-right corner of the AWS Console, next to your username. For example:

- *United States(N. California)* → us-west-1
- *United States(Oregon)* → us-west-2
- *United States(N. Virginia)* → us-east-1
- *United States(Ohio)* → us-east-2

All the instances and resources you create are tied to the selected **region**. The next time you log in, make sure you are in the **same region** so you can access your previously created resources. If not, use the dropdown menu to switch to the correct region.

Step 3. Go to EC2 dashboard (search "EC2" if you cannot find it). The current default is the 'New EC2 console'. Click the "Launch Instance" button to start.

Step 4. In the "Name and tags", fill in a name for your instance (for example, "Homework1")

Step 5, In the "Application and OS Images (Amazon Machine Image) ", explore all the different AMI(Amazon Machine Images) listed there. Under the "Quick Start" keep default "Amazon Linux".

Step 6. choose instance type, use pull down menu, select "t3.micro"

Step 7. In "Key pair(login)", select "create new key pair". In the pop-up dialog box, enter a name name(you can use anything string, for example, 'SCU_2025Fall'), then click "Create key pair". A file with .pem extension will be downloaded into your local computer download folder. You will need this key to connect to your AWS instance.

Step 8. In “Network settings”, enable “Allow HTTP traffic from the internet”

Step 9. In the right side “Summary”, click “Launch Instance”

Step 10. If everything correct, then your first instance will be launched successfully.

- It will be a good idea to use the “Create billing and free tier usage alerts” to configure your free tier usage alerts.

Step 11. Go back to the EC2 dashboard, pinpoint the running instance just created, find out "Public DNS" from its description (e.g. **ec2-1-2-34-56.us-east-1.compute.amazonaws.com**)

Step 12. Go to your local Unix terminal(assuming you are in your home directory like: /Users/your_username, copy the keypairs file '**SCU_2025Fall**' into your current directory) then connect to the EC2 instance using the following commands:

```
$ chmod 400 SCU_2025Fall.pem
$ ssh -i SCU_2025Fall ec2-user@ec2-1-2-34-56.us-east-1.compute.amazonaws.com (replace
the correct EC2 instance name)
```

If you are using **Windows**, either choose Windows Subsystem for Linux (WSL) or install **putty** referring to <https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/connect-linux-inst-from-windows.html>

Step 13. In the EC2 instance, in your current home directory (/home/ec2-user), with **vi** editor (or '**nano**' editor) and create a simple html file named as **index.html**.

```
[ec2-user@....]$ vi index.html
```

The sample content of index.html is as follows:

```
<html>
<body>
<p> *** Hello SCU! *** </p>
</body>
</html>
```

Hint: use [ec2-user@....]\$ **more index.html** command to check the file content)

Step 14, Now change directory to directory /var/www/html and copy the index.html file to this directory:

Note: If the /var/www and /var/www/html directory doesn't exist, then you can create them by using:

```
[ec2-user@....]$ sudo mkdir /var/www
[ec2-user@....]$ sudo mkdir /var/www/html
```

Then change directory to /var/www/html by

```
[ec2-user@....]$ cd /var/www/html
```

```
[ec2-user@....]$ sudo cp ~/index.html .
```

Step 15. Install Apache web server by:

```
[ec2-user@....]$ sudo yum install -y httpd
```

Step 16. Launch a http server using the command:

```
[ec2-user@....]$ sudo systemctl start httpd
```

Step 17. In your local computer browser, check whether your http server is working or not by typing `http://your_ec2_instance_name` for example:

`http://ec2-1-2-34-56.us-east-1.compute.amazonaws.com`

Step 18. Now expand your index.html a little bit by adding other type of tags/content (no need to make it too complex)

Step 19: Start to answer all questions in next page(**Assignment Deliverables**).

Step 20: When all questions are answered, (or you have to leave for few hours), you need to **shut down** all EC2 instance(s) by following steps:

In EC2 Dash board -> Instances (running), select the running instance, then on upper right corner "Instance state" pull down menu, select "stop instance".

- **Do NOT leave your instance(s) free run overnight or several hours without actually using it. It will waste your free hours(if any) or cause un-necessary charges into your credit card.**
- **From time to time, check the "AWS Billing Dashboard " and make sure you are not being charged by your free running instance(s).**

Assignment Deliverables

Part 1: (50 pts)

Document the EC2 launch process that includes:

1. Screenshot of the EC2 instance description which shows instance state, Instance type, Availability Zone, Security Groups, Public DNS. (5 pts)

The screenshot displays the AWS Management Console interface for EC2 instances. At the top, there's a header with 'Instances (1/4)' and a search bar. Below this is a table listing instances. The instance 'Homework1' with ID 'i-0d7d201b7a009ae4d' is selected. Below the table, the detailed view for this instance is shown, including its state (Stopped), type (t3.micro), and various addresses. A second, larger screenshot below shows the 'Instance summary' for the same instance, which is now in a 'Running' state. This summary provides more details, including the public IPv4 address (13.57.218.54), private IP DNS name, VPC ID, Subnet ID, and Instance ARN.

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone
Homework1	i-0d7d201b7a009ae4d	Stopped	t3.micro	-	View alarms +	us-west-1a
HW1Part2Instance_2	i-024fcf6186fb7c347	Running	t3.micro	3/3 checks passed	View alarms +	us-west-1a
A	i-089b7d9cb9dd9aca	Stopped	t3.micro	-	View alarms +	us-west-1a
HW1Part2Instance_1	i-0b8a37974176c06c9	Running	t3.micro	3/3 checks passed	View alarms +	us-west-1a

i-0d7d201b7a009ae4d (Homework1)

Details | Status and alarms | Monitoring | Security | Networking | Storage | Tags

Instance summary Info

Instance ID: i-0d7d201b7a009ae4d

IPv6 address: -

Hostname type: IP name: ip-172-31-20-237.us-west-1.compute.internal

Answer private resource DNS name: IPv4 (A)

Auto-assigned IP address: -

Public IPv4 address: -

Instance state: Stopped

Private IP DNS name (IPv4 only): ip-172-31-20-237.us-west-1.compute.internal

Instance type: t3.micro

VPC ID: vpc-091ffc5ef10a67c8f

Private IPv4 addresses: 172.31.20.237

Public DNS: -

Elastic IP addresses: -

AWS Compute Optimizer finding: Opt-in to AWS Compute Optimizer for recommendations. | Learn more

Instance summary for i-0d7d201b7a009ae4d (Homework1) Info

Updated 36 minutes ago

Instance ID: i-0d7d201b7a009ae4d

IPv6 address: -

Hostname type: IP name: ip-172-31-20-237.us-west-1.compute.internal

Answer private resource DNS name: IPv4 (A)

Auto-assigned IP address: 13.57.218.54 [Public IP]

IAM Role: -

IMDSv2: Required

Operator: -

Public IPv4 address: 13.57.218.54 | open address

Instance state: Running

Private IP DNS name (IPv4 only): ip-172-31-20-237.us-west-1.compute.internal

Instance type: t3.micro

VPC ID: vpc-091ffc5ef10a67c8f

Subnet ID: subnet-05dca873c49f84a59

Instance ARN: arn:aws:ec2:us-west-1:293578648063:instance/i-0d7d201b7a009ae4d

Private IPv4 addresses: 172.31.20.237

Public DNS: ec2-13-57-218-54.us-west-1.compute.amazonaws.com | open address

Elastic IP addresses: -

AWS Compute Optimizer finding: Opt-in to AWS Compute Optimizer for recommendations. | Learn more

Auto Scaling Group name: -

Managed: false

Details	Status and alarms	Monitoring	Security	Networking	Storage	Tags
VPC ID vpc-091ffc5ef10a67c8f			Subnet ID subnet-05dca873c49f84a59		Availability zone us-west-1a	
Availability zone ID usw1-az1			Outpost ID -			
▼ IP addresses Info						
Public IPv4 address 13.57.218.54 open address			Private IPv4 addresses 172.31.20.237		IPv6 addresses -	
Secondary private IPv4 addresses -			Carrier IP addresses (ephemeral) -			
▼ Hostname and DNS Info						
Public DNS ec2-13-57-218-54.us-west-1.compute.amazonaws.com open address			Private IP DNS name (IPv4 only) ip-172-31-20-237.us-west-1.compute.internal		IPv4-only IP based name: A record only ec2-13-57-218-54.us-west-1.compute.amazonaws.com	
Dualstack - IP based name: A and AAAA record -			IPv6-only - IP based name: AAAA record only -		Public hostname type public-ipv4-dns-name	
Private hostname type IP name: ip-172-31-20-237.us-west-1.compute.internal			Use RBN as guest OS hostname Disabled		Answer RBN DNS hostname IPv4 Enabled	
Answer RBN DNS hostname IPv6 -			Answer private resource DNS name IPv4 (A)			

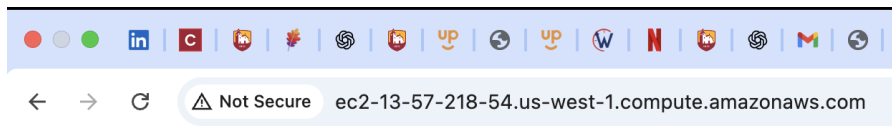
Details	Status and alarms	Monitoring	Security	Networking	Storage	Tags
▼ Security details						
IAM Role		Owner ID			Launch time	
-		 293578648063			Sun Sep 28 2025 00:17:34 GMT-0700 (Pacific Daylight Time)	
Security groups						
 sg-0f063f8721fcc87fa (launch-wizard-2)						

Details	Status and alarms	Monitoring	Security	Networking	Storage	Tags
▼ Instance details Info						
AMI ID ami-00142eb1747a493d9		Monitoring disabled		Platform details Linux/UNIX		
AMI name al2023-ami-2023.8.20250915.0-kernel-6.1-x86_64		Allowed image -		Termination protection Disabled		
Stop protection Disabled		Launch time Sun Sep 28 2025 00:17:34 GMT-0700 (Pacific Daylight Time) (38 minutes)		AMI location amazon/al2023-ami-2023.8.20250915.0-kernel-6.1-x86_64		
Instance reboot migration Default (On)		Instance auto-recovery Default		Lifecycle normal		
Stop-hibernate behavior Disabled		AMI Launch index 0		Key pair assigned at launch SCU_2025Fall		
State transition reason -		Credit specification unlimited		Kernel ID -		
State transition message -		Usage operation RunInstances		RAM disk ID -		
Owner 293578648063		Enclaves Support Disabled		Boot mode uefi-preferred		
Current instance boot mode uefi		Allow tags in instance metadata Disabled		Use RBN as guest OS hostname Disabled		
Answer RBN DNS hostname IPv4 Enabled						

2. Screenshot of the welcome page after your SSH connect to the EC2 instance. (5 pts)

[illegible]

- 
- The screenshot shows a web browser interface. The address bar displays the URL 'ec2-13-57-218-54.us-west-2.amazonaws.com'. A yellow warning icon and the text 'Not Secure' are visible in the address bar. The page content shows a large, bold, black text '***Hello SCU!***'.



Here is a simple list of languages:

5. Explain what is security group. We created a new security group that consists of SSH and HTTP rule. What are the default ports for SSH and HTTP in respect? Is the HTTP rule a

must in this assignment? Can you “ping” the AWS instance from your local host and explain why. (5 pts)

Answer - A security group in AWS works like a virtual firewall for your EC2 instance. It decides which inbound and outbound network traffic is allowed. By default, all inbound traffic is blocked, so you have to add rules to let specific connections (like SSH or HTTP) through. Outbound traffic is generally allowed unless you restrict it.

The default port for SSH is 22, which is used to securely connect to the instance from your terminal. The default port for HTTP is 80, which is the standard for serving web pages to browsers. These ports are well-known and must be opened in the security group to allow those types of connections.

Yes, the HTTP rule is required because you need to access the webpage (index.html) hosted by Apache. Without port 80 open, your browser won't be able to reach the server even though the instance is running.

By default, you cannot ping an AWS instance from your local machine. That's because ping uses the ICMP protocol, and security groups don't allow ICMP unless you add a specific rule for it. This restriction helps protect your instance from unnecessary network probing.

6. What is availability zone? What availability zone do you choose for your ec2 instance and why? (5 pts)

An availability zone is an isolated data center within an AWS region that has its own power, networking, and cooling. AWS regions are made up of multiple availability zones, and they are designed so that if one availability zone goes down, others in the region can still function. This provides high availability and fault tolerance for applications. I chose the availability zone in my default region (us-west-1a in Northern California) since it was closest to me and already selected in the AWS console. Using the nearest availability zone usually gives lower latency when I connect from my local machine, and it also keeps all my resources in the same region to avoid confusion or extra costs.

7. We change the key pair file mode using the command "`chmod 400 SCU_2025Fall.pem`". What is the meaning of 400? What if we do not change the file mode? (5 pts)

The number 400 in `chmod 400` sets the file permission so that only the file's owner can read it. The first digit 4 means “read” permission, and the two zeros mean no write or execute permissions for the owner, group, or others. This ensures the private key file is read-only and secure.

If the .pem file is left with more open permissions (like writable or accessible to others), SSH will refuse to use it because it's considered insecure. In that case, you'll get a “bad permissions” or “unprotected private key file” error, and you won't be able to connect to your EC2 instance. Setting it to 400 prevents that problem.

8. If you are qualify to use AWS free tier instances, how many free hours do you have in total in the first 12 months? Explain what is AMI at step 4? (5 pts)

If you qualify for AWS Free Tier, you get 750 hours per month of t3.micro instances. Over 12 months, that equals 9,000 free hours in total. This is enough to run one instance continuously each month.

An Amazon Machine image (AMI) is a template that contains the operating system, application server, and any software needed to launch an EC2 instance. At Step 4, I selected an AMI, which was Amazon Linux by default, which defined what the instance will look like once it is launched.

9. In step 16 why do we need to use sudo ? If you are not allowed to run the http server with sudo, but still you want to browse your web page in the browser, what should you do? (5 ps)

The command `sudo systemctl start httpd` is used because starting or stopping system services like Apache requires administrator (root) privileges. Without `sudo`, a regular user does not have permission to manage system-level daemons.

If you are not allowed to use `sudo`, you could run a web server on a non-privileged port (above 1024) that doesn't require root access, such as running Python's simple HTTP server on port 8080. You would then access your page in the browser using `http://your-ec2-dns:8080` instead of port 80.

10. Please download the syllabus file of this course to your local system then figure out yourself how to transfer that file to your EC2 instance. Capture two screenshots:

- One on your local system showing the exact command you used to perform the transfer.

```
(base) shwetaleegadekar@Shwetalees-MacBook-Air-1956 ~ % cp ./Downloads/ISBA2416_Fall12025_AppliedCloud_TTh735.pdf .
(base) shwetaleegadekar@Shwetalees-MacBook-Air-1956 ~ % ls
Desktop                               Movies                                Public
Documents                             Music                                PycharmProjects
Downloads                             nltk_data                            SCU_2025Fall.pem
ISBA2416_Fall12025_AppliedCloud_TTh735.pdf  Parallels                           Untitled.ipynb
Library                               Pictures                             Virtual Machines.localized
(base) shwetaleegadekar@Shwetalees-MacBook-Air-1956 ~ % scp -i SCU_2025Fall.pem ISBA2416_Fall12025_AppliedCloud_TTh735.pdf ec2-user@ec2-13-57-218-54.us-west-1.compute.amazonaws.com:/home/ec2-user/
ISBA2416_Fall12025_AppliedCloud_TTh735.pdf                                100% 177KB  1.5MB/s  00:00
```

- One on your EC2 instance confirming that the file was successfully copied. (5 pts)

```
(base) shweta.leegadekar@ShwetaLees-MacBook-Air-1956 ~ % ssh -i SCU_2025Fall.pem ec2-user@ec2-13-57-218-54.us-west-1.compute.amazonaws.com
```

```
_#_
~\####_ Amazon Linux 2023
~~~\#####\
~~~~\###|
~~~~\#/
~~~~V'---> https://aws.amazon.com/linux/amazon-linux-2023
~~~~
~~~~_. _/
~~~~/_/
~~~~/_m/'
```

```
Last login: Sun Sep 28 07:33:33 2025 from 98.42.40.92
[ec2-user@ip-172-31-20-237 ~]$ ls -l ISBA2416_Fall2025_AppliedCloud_TTh735.pdf
-rw-r--r--. 1 ec2-user ec2-user 181454 Sep 28 08:33 ISBA2416_Fall2025_AppliedCloud_TTh735.pdf
```

This is end of Part 1. Don't forget to stop your instance when all questions are answered !